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Alpha-Mannosidosis

Synonym: α-Mannosidosis

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Summary

Clinical characteristics

The clinical phenotype of alpha-mannosidosis varies considerably, with a wide spectrum of clinical findings and broad variability in individual presentation. At least three clinical types have been suggested in untreated individuals: mild (clinically recognized after age ten years, with myopathy, slow progression, and absence of skeletal abnormalities); moderate (clinically recognized before age ten years, with myopathy, slow progression, and presence of skeletal abnormalities); and severe (obvious progression leading to early death from primary central nervous system involvement or infection). Core features of untreated individuals generally include early childhood-onset non-progressive hearing loss, frequent infections due to immunodeficiency, rheumatologic symptoms (especially systemic lupus erythematosus), developmental delay / intellectual disability, low tone, ataxia, spastic paraplegia, psychiatric findings, bone disease (ranging from asymptomatic osteopenia to focal lytic or sclerotic lesions and osteonecrosis), gastrointestinal dysfunction (including diarrhea, swallowing issues / aspiration, and enlarged liver and spleen), poor growth, eye issues (including tapetoretinal degeneration and optic nerve atrophy), cardiac complications in adults, and pulmonary issues (including parenchymal lung disease). However, with the advent of enzyme replacement therapy, the natural history of this condition may change. Long-term velmanase alfa (VA) treatment outcomes are still being elucidated, but may include improvement in hearing, immunologic profile, and quality of life (improved clinical outcomes for muscle strength). Similarly, affected individuals who underwent hematopoietic stem cell transplantation (HSCT) experienced improvement in development (with preservation of previously learned skills), ability to participate in activities of daily living, stabilization or improvement in skeletal abnormalities, and improvement in hearing ability, although expressive speech and hearing deficiencies remained the most significant clinical problems after HSCT.

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Diagnosis/testing

The diagnosis of alpha-mannosidosis is established in a proband by identification of deficiency of lysosomal enzyme acid alpha-mannosidase (typically 5%-10% of normal activity) in leukocytes or other nucleated cells AND/OR by the identification of biallelic pathogenic variants in *MAN2B1* by molecular genetic testing.

Management

Targeted therapies: Velmanase alfa (Lamzede®) enzyme replacement therapy (ERT) has been very well tolerated and is now regarded as a standard treatment for alpha-mannosidosis; improvement in both biochemical and functional parameters have been reported in treated individuals. Hematopoietic stem cell transplantation (HSCT) has been offered as a treatment for severe alpha-mannosidosis. While HSCT carries risks, the data suggests it is a feasible therapeutic option for alpha-mannosidosis, with better outcomes achieved by performing it early before complications arise, balancing the risks and benefits.

Supportive care: Hearing aids may be helpful for those with sensorineural hearing loss, whereas pressure-equalizing tubes may be helpful for those with conductive hearing loss. Consider palmidronate (Aredia®) monthly or zoledronic acid (Aclasta®) once a year for osteoporosis or osteopenia. Standard treatment for immunodeficiency / recurrent infections, systemic lupus erythematosus, communicating hydrocephalus, ataxia / gait abnormalities, poor weight gain / growth issues, eye/vision issues, cardiac valve dysfunction / dilated cardiomyopathy, recurrent chest infections / respiratory dysfunction, developmental delay / intellectual disability, and psychiatric manifestations.

Surveillance: At each visit, measure weight, length/height, head circumference, and BMI; monitor growth pattern, developmental progress, and educational needs; assess for depression, including sleep disturbances, anxiety, &/or findings suggestive of psychosis; assess for new manifestations such as ataxia and gait abnormalities; evaluate for asthenia and signs/symptoms of communicating hydrocephalus; assess for muscle pain, joint aches, reduced range of motion, and bone pain; monitor for diarrhea and for size of the liver and spleen; and assess for the number and type of infections. Every six to 12 months in childhood and annually in adults, assess fine motor function, gross motor function, endurance, and muscle strength and tone by physical therapy; and assess for features of ataxia. Every one to two years, or as clinically indicated in those with hearing aids, perform an audiology evaluation. Every two to five years in children, adolescents, and adults, consider DXA bone densitometry scan to assess for osteopenia or osteoporosis; radiographs of the hips/spine may be indicated. Annually (or as clinically indicated), routine biochemical lab assessment to include liver and kidney health, blood glucose levels, fluid and electrolyte balance, and complete blood count (with platelets); consider immunlobulin levels, ESR and C-reactive protein; pulmonary function tests; and ophthalmology evaluation. At regular intervals based on clinical features, consider endocrinology evaluations, including hormonal and lipid profiles; consider assessment of liver and spleen size through ultrasound or MRI imaging; consider electrocardiogram, 24-hour electrocardiogram, and echocardiogram; consider sleep study. For those on ERT, plasma oligosaccharides to assess treatment response as clinically indicated. Post-HSCT evaluation of standard surveillance per hematologist/oncologist, which may include ongoing assessment for chimerism and enzyme activity if indicated.

Evaluation of relatives at risk: Testing of all at-risk sibs of any age (including prenatal diagnosis) is warranted to allow for early diagnosis and targeted treatment of alpha-mannosidosis. Evaluations can include molecular genetic testing if the pathogenic variants in the family are known or assay of acid alpha-mannosidase enzyme activity in leukocytes or other nucleated cells if the pathogenic variants in the family are not known.

Differential Diagnosis

Lysosomal storage disorders. The main clinical features in alpha-mannosidosis – intellectual disability, ataxia, coarse face, and dysostosis multiplex – may overlap with other lysosomal storage disorders (e.g., mucopolysaccharidosis type I and II). However, the distinctive clinical features associated with other lysosomal storage disorders, the availability of biochemical testing in clinical laboratories, and an understanding of their natural history should help in distinguishing between them.

Table 2. Genes of Interest in the Differential Diagnosis of Alpha-Mannosidosis

Gene(s)	Disorder	MOI	Key Clinical Features of Disorder	
			Overlapping w/alpha-mannosidosis	Distinguishing from alpha-mannosidosis
ABCC9 *KCNJ8*	Cantú syndrome	AD	• Coarse facial features • Thickened ribs	• Heart defects • Hypertrichosis
ARSB *ARSK* *GALNS* *GLB1* *GNS* *GUSB* *HGSNAT* *HYAL1* *IDS* *IDUA* *NAGLU* *SGSH*	Mucopolysaccharidoses (OMIM PS607014)	AR XL [1]	• Coarse facial features • Dysostosis multiplex • ID	• Short stature • Contractures
GNE	Sialuria (OMIM 269921)	AD	• Hypotonia • Coarse facial features • DD • Frequent upper respiratory infections	• Joint stiffness • Seizures • Microcytic anemia
GNPTAB	Mucolipidosis II (See *GNPTAB*-Related Disorders.)	AR	• Coarse facial features • Dysostosis multiplex	• Short stature • Failure to thrive
	Mucolipidosis IIIα/β (See *GNPTAB*-Related Disorders.)			• Short stature • Normal-to-mildly impaired cognitive development
NEU1	Sialidosis (OMIM 256550)	AR	• Coarse facial features • Dysostosis multiplex • ID	Cherry-red spot of the macula
SUMF1	Multiple sulfatase deficiency	AR	• Similar to MPS II • DD	• Poor feeding • Retinopathy

AD = autosomal dominant; AR = autosomal recessive; DD = developmental delay; ID = intellectual disability; MOI = mode of inheritance; XL = X-linked

1. The mucopolysaccharidoses are inherited in an autosomal recessive manner with the exception of mucopolysaccharidosis type II, which is associated with pathogenic variants in *IDS* and inherited in an X-linked manner.

Management

No clinical practice guidelines for alpha-mannosidosis have been published.

Table 3. continued from previous page.

System/Concern	Evaluation	Comment
Eyes	Ophthalmologic eval	To assess for reduced vision, best corrected visual acuity, refractive errors, strabismus, & more complex findings (e.g., corneal opacities, retinal dystrophy) that may require referral for subspecialty care &/or low vision services
Cardiovascular	Auscultation w/consideration of echocardiogram	• To assess for aortic or mitral stenosis/regurgitation &/or cardiomyopathy • Consider referral to cardiologist.
Respiratory	Baseline pulmonary function tests in those older than age 6 yrs	Consider referral to pulmonologist.
Genetic counseling	By genetics professionals [3]	To obtain a pedigree & inform affected persons & their families re nature, MOI, & implications of alpha-mannosidosis to facilitate medical & personal decision making
Family support & resources	By clinicians, wider care team, & family support organizations	Assessment of family & social structure to determine need for: • Community or online resources such as Parent to Parent • Social work involvement for parental support • Home nursing referral

Based on Ficicioglu et al [2024]
ADL = activities of daily living; DXA = dual-energy x-ray absorptiometry; MOI = mode of inheritance; OT = occupational therapy; PT = physical therapy; SLE = systemic lupus erythematosus
1. Such as change in social, domestic, or school- or work-related activities or in ability to walk distances
2. Including headache, increasing gait ataxia, nausea, and/or papilledema
3. Medical geneticist, certified genetic counselor, certified advanced genetic nurse

Treatment of Manifestations

There is no cure for alpha-mannosidosis.

Targeted Therapies

In GeneReviews, a targeted therapy is one that addresses the specific underlying mechanism of disease causation (regardless of whether the therapy is significantly efficacious for one or more manifestation of the genetic condition); would otherwise not be considered without knowledge of the underlying genetic cause of the condition; or could lead to a cure. —ED

Velmanase alfa (Lamzede®) enzyme replacement therapy (ERT) has been approved by the FDA for the treatment of the non-central nervous system features of alpha-mannosidosis.

Table 4. Alpha-Mannosidosis: Targeted Treatment

Targeted Treatment	Dosage	Consideration
Velmanase alfa (Lamzede®) ERT	1 mg/kg IV infusion once a week [1]	• Improvement in both biochemical & functional parameters have been reported. [2] • Velmanase alfa has been very well tolerated & is now regarded as standard treatment for alpha-mannosidosis.

ERT = enzyme replacement therapy
1. For individuals who weigh less than 49 kg, IV infusion is typically given over a minimum of 60 minutes; for those who weigh 50 kg or more, infusion rate is typically 25 mL/hour.
2. Borgwardt et al [2018], Lund et al [2018]

Evaluations Following Initial Diagnosis

To establish the extent of disease and needs in an individual diagnosed with alpha-mannosidosis, the evaluations summarized in Table 3 (if not performed as part of the evaluation that led to the diagnosis) are recommended.

Table 3. Alpha-Mannosidosis: Recommended Evaluations Following Initial Diagnosis

System/Concern	Evaluation	Comment
Constitutional	Measurement of weight, length/height, & head circumference	To assess for poor growth &/or macrocephaly
Hearing	Audiologic eval & assessment for middle ear effusions	Assess for both sensorineural & conductive hearing loss.
Immunologic/ Rheumatologic	Assess for signs/symptoms of frequent infections.	Consider referral to immunologist.
	Clinical & laboratory assessment for features of SLE	• To incl immunologic testing such as anti-nuclear antibodies & anti-double-stranded-DNA antibodies • Consider referral to rheumatologist.
Development	Developmental assessment	• To incl motor, adaptive, cognitive, & speech-language eval • Eval for early intervention / special education
Neurobehavioral/ Psychiatric	Neuropsychiatric eval	For persons age >12 mos: screening for concerns incl depression, sleep disturbances, anxiety, &/or findings suggestive of psychosis (delusions, hallucinations)
Neurologic	Neurologic eval	• Assess for asthenia [1] & signs/symptoms of communicating hydrocephalus. [2] • Consider head CT to assess size of ventricles & shape/size of cerebellum, particularly if signs/symptoms of hydrocephalus are present. • Assess for signs/symptoms of ataxia & gait abnormalities.
Musculoskeletal	Orthopedics / physical medicine & rehab / PT & OT eval	To incl assessment of: • Gross motor & fine motor skills • Muscle pain, joint aches, reduced range of motion, & bone pain • Mobility, ADL, & need for adaptive devices • Need for PT (to improve gross motor skills) &/or OT (to improve fine motor skills)
	Plain radiographs of head, knees (AP view), spine (lateral view), & any symptomatic sites	To assess for lytic bone lesions
	Consider DXA bone densitometry scan to assess for osteopenia or osteoporosis.	In older children, adolescents, & adults
Gastrointestinal/ Feeding	Gastroenterology / nutrition / feeding team eval	• To incl eval of aspiration risk & nutritional status • Consider eval for gastrostomy tube placement in persons w/ dysphagia &/or aspiration risk.
	Assess for evidence of diarrhea & for enlarged liver &/or spleen via physical exam & imaging	• Ultrasound is often done, although abdominal MRI may be more informative to calculate organ volumes. • Although organomegaly typically does not cause clinical symptoms, this finding can be used as a clinical marker to assess treatment response.